Quantitative Economics

Spring 2027 — Academic year 2026/2027
University of Warsaw

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Course Description This course introduces numerical methods and dynamic programming for economic analysis. We will study consumption and saving, income dynamics and wealth distributions in partial and general equilibrium, and solve the resulting models numerically.

The course combines economic theory, mathematical reasoning and programming. Topics include Bellman equations, numerical solution methods and the interpretation of model results.

We will use Julia. No prior knowledge of the language is required; an introduction and example code will be provided.

Prerequisites: linear algebra, mathematical analysis and probability.

Outline

1. Programming in Julia.
2. Efficient and reproducible workflow: version control, debugging, benchmarking, modules and packages.
3. Elementary concepts in scientific computing.
4. Linear and nonlinear equations and systems.
5. Optimization.
6. Function approximation and interpolation.
7. Numerical integration and Markov processes.
8. Introduction to dynamic programming.
9. Discrete Markov decision processes.
10. Continuous-state models.
11. Heterogeneous agent models: Bewley–Huggett–Aiyagari economies.

Assessment The assessment weights are:

- Final written examination: 55%.
- Two short in-class tests: 15% in total.
- Three problem sets: 30% in total.

The problem sets will involve theoretical derivations and computational applications of material covered in class. You may work in groups of up to five students and discuss solutions with students outside your group. Credit everyone with whom you worked. You will have two weeks for each problem set. Submit your Julia code together with a write-up.

There will be two short in-class tests, completed individually. Each is expected to last 15–20 minutes and will check your understanding of the basics of programming in Julia.

The final assessment is an individual, closed-book written examination held in person.

On Using AI AI tools are permitted as learning aids for problem sets, but your submissions must demonstrate your own understanding and meaningful contribution. Acceptable uses include getting help with debugging, learning new syntax, and reviewing code you have written. You are responsible for checking the correctness of all submitted code and results and must be able to explain your methods. Do not submit solutions you cannot explain. If we have questions about your approach or understanding, we may ask you to explain your solutions in a brief meeting. The goal is to develop your ability to formulate economic problems, choose numerical methods and evaluate the results.

Materials There is no single textbook for this course. We will mostly rely on materials posted in the course repository. The following books provide background on the theoretical aspects of the course:

- Stokey, Lucas and Prescott (1989), *Recursive Methods in Economic Dynamics*.
- Ljungqvist and Sargent (2018), *Recursive Macroeconomic Theory*.
- Sargent and Stachurski (2025), *Dynamic Programming, Volume I: Finite States* (online version and code).
- Jappelli and Pistaferri (2017), *The Economics of Consumption*.

For an introduction to Julia:

- Kamiński (2022), *Julia for Data Analysis*.

Additional resources on computation in economics:

- Quantitative Economics with Julia from QuantEcon. Relevant lectures include the McCall job search model, discrete state dynamic programming, the endogenous grid method, and the Aiyagari model.
- Heer and Maussner (2009), *Dynamic General Equilibrium Modeling*.
- Miranda and Fackler (2002), *Applied Computational Economics and Finance*.
- Judd (1998), *Numerical Methods in Economics*.

For more detail on numerical analysis:

- Cheney and Kincaid (2008), *Numerical Mathematics and Computing*, 6th edition.